Introduction to Artificial Intelligence

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01

What is Artificial Intelligence

Before we dive into the details of data science and AI development, let's first discuss a fundamental question - what is artificial intelligence?

What is Intelligence?

General Definition: Is the capacity to acquire, understand, and use knowledge and skills to adapt to new circumstances.

Cognitive Perspective: Mental ability to reason, solve problems, comprehend complex ideas and learn from experience.

Emotional Intelligence: Ability to recognize, understand, and manage emotions.

Social Intelligence: Is the capability to navigate complex social environments, understanding social cues, and forming social bonds.

Biological Perspective: Is a product of brain activity, enabling an organism to process information from the environment, make decisions, and engage in complex behaviors.

Philosophical View: Is the ability to think critically, question assumptions, and shape the world in meaningful ways.

Practical Intelligence: Intelligence is the ability to solve real-world problems, adapt to changing environments, and effectively navigate everyday challenges and opportunities.

Many different terms and descriptions from different perspectives

* All the terms are loosely quoted from various sources, shortened for slide simplification.

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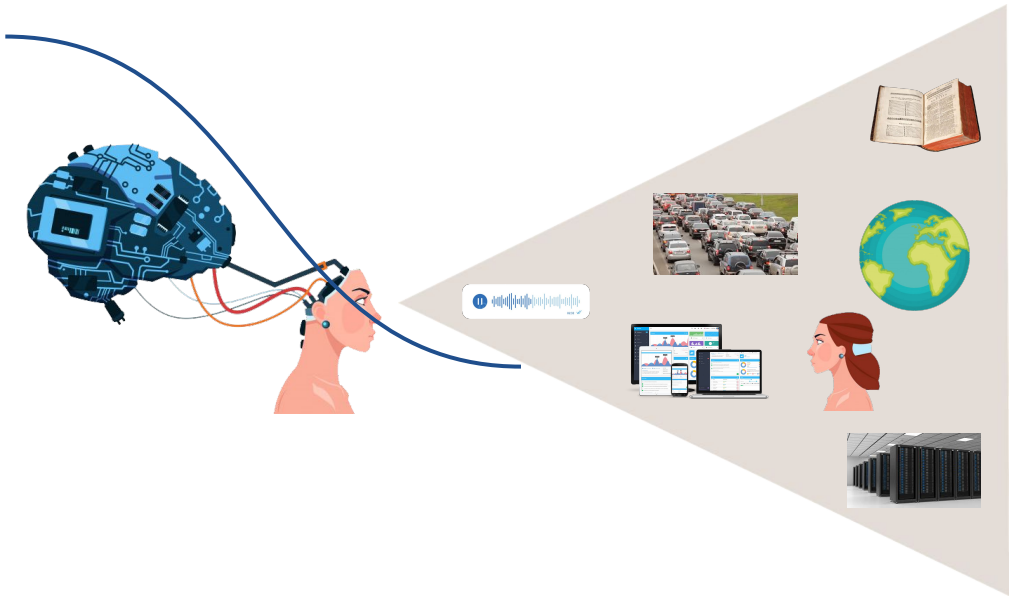
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Practical Intelligence: Intelligence is the ability to solve real-world problems, adapt to changing environments, and effectively navigate everyday challenges and opportunities.

COMMUNICATION UNDERSTANDING ADAPTATION

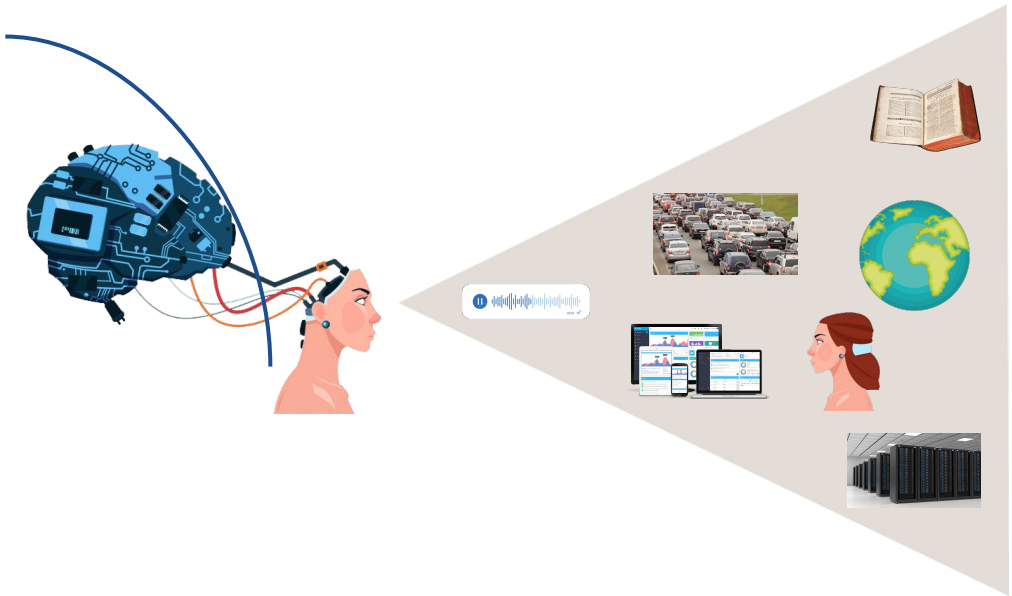
Artificial Intelligence

COMMUNICATION: This part of AI addresses questions about how we collect data, how we store it for AI purposes, what business logic underlies our decisions, and how they impact the world. This area overlaps with fields such as sensor engineering, big data management, ethics, and more.



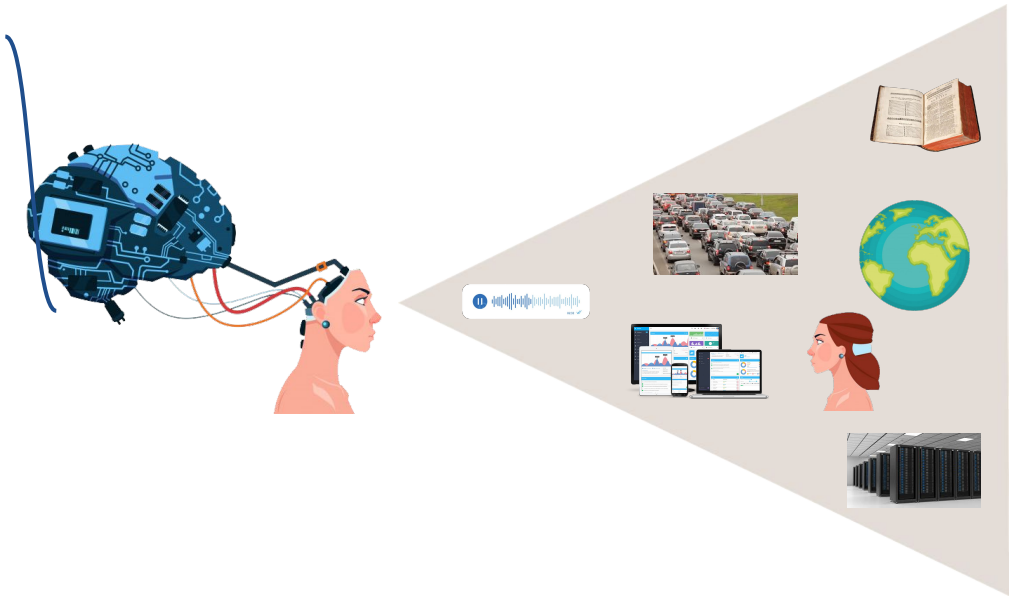
Artificial Intelligence

UNDERSTANDING: This area of AI can be conditionally designated for addressing issues related to data preprocessing, aimed at optimizing the data for subsequent processing by more advanced algorithms. Additionally, this section intersects with technical aspects of data delivery and ETL processes.



Artificial Intelligence

ADAPTATION: This is a key part of AI that we will be focusing on and exploring. It includes questions about how to represent information, learn, and use accumulated knowledge to solve previously unseen problems. Specifically, in this section, we will be discussing machine learning extensively.



02

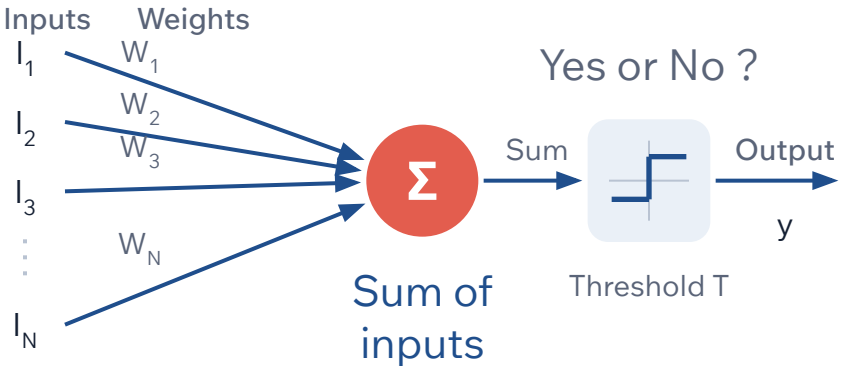
History of Artificial Intelligence

Now when we've explored the components of artificial intelligence, let's examine its origins and development process.

Origins of Artificial Intelligence



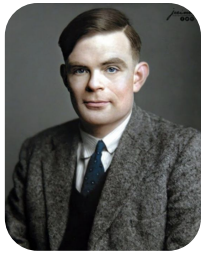
Walter Pitts (mathematician)
Warren McCulloch (neurophysiologist)



1943: First formal Artificial NN.

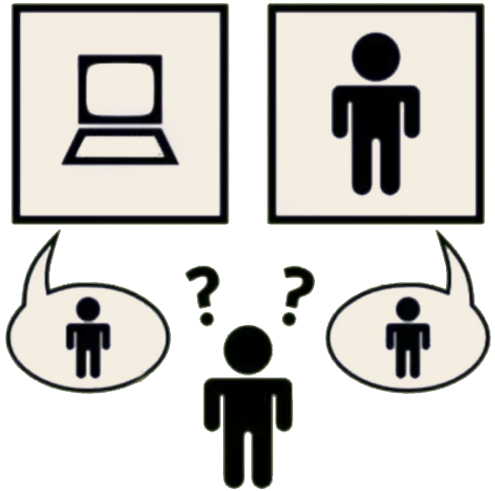
Just before the AI was originated as a field of study.

Origins of Artificial Intelligence

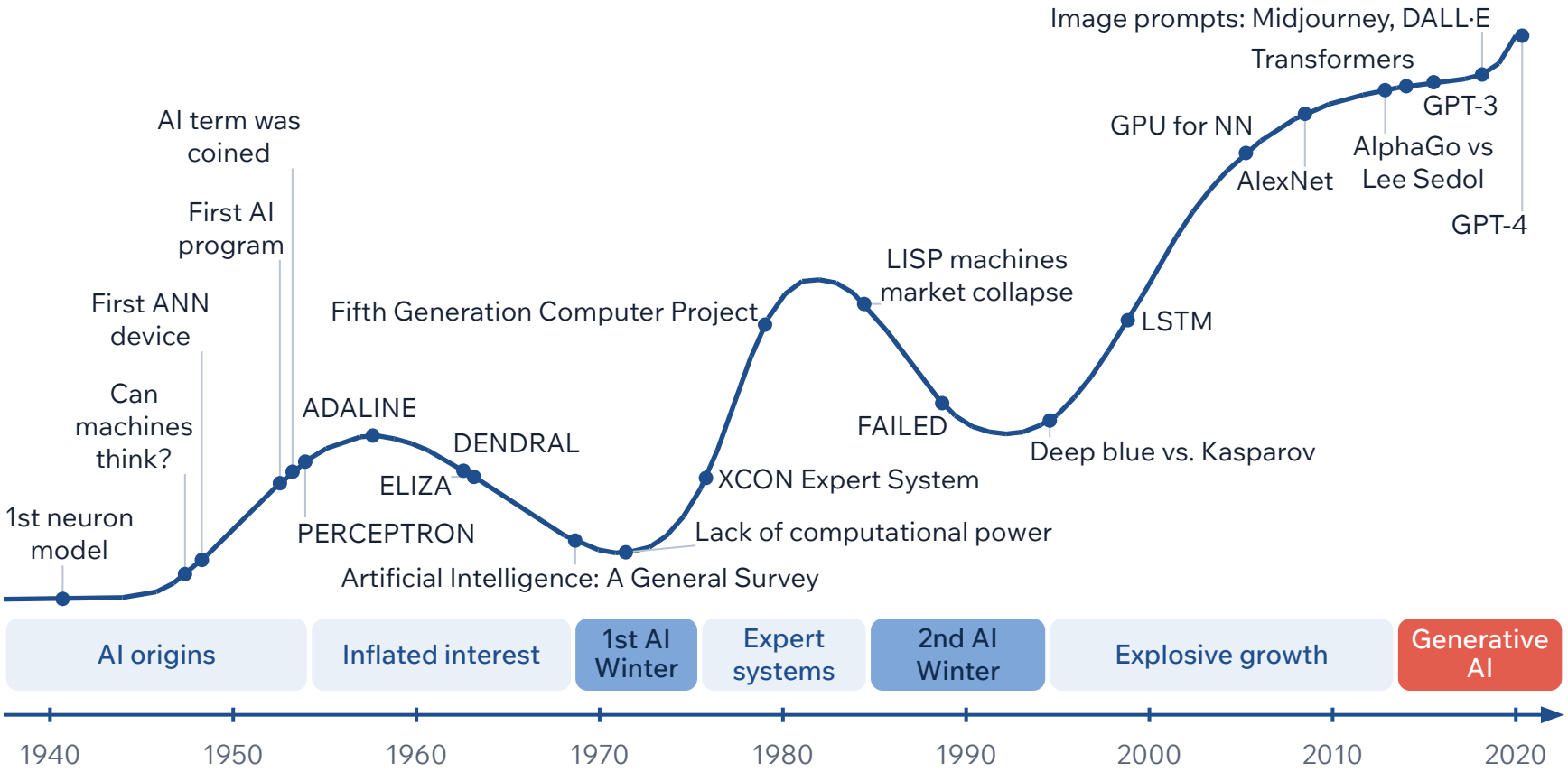


Alan Turing (mathematician)
1950: «Computing Machinery and intelligence»

Alan Turing introduced the concept known as the “Turing Test,” although he did not intend to create a test. His goal was to explore whether a machine could exhibit behavior similar to human intelligence, using an imitation game to discuss the possibility of machine thinking.



Person (judge) communicates via text with two participants, one of whom is a human and the other a machine. If the judge cannot confidently determine which one is the machine, it is considered that the machine has successfully imitated human intelligence.


Image prompts: Midjourney, DALL·E
Transformers
GPT-3
GPU for NN
AlphaGo vs
Lee Sedol
AlexNet
GPT-4
AI term was
coined
First AI
program
First ANN
device
LISP machines
market collapse
Fifth Generation Computer Project
LSTM
Can
machines
think?
ADALINE
DENDRAL
FAILED
ELIZA
XCON Expert System
Deep blue vs. Kasparov
1st neuron
model
PERCEPTRON
Lack of computational power
Artificial Intelligence: A General Survey
AI origins
Inflated interest
1st AI
Winter
Expert
systems
2nd AI
Winter
Explosive growth
Generative
AI
1940
1950
1960
1970
1980
1990
2000
2010
2020

AI origins

- 1st neuron model
- Can machines think?
- First ANN device
- First AI program
- AI term was coined
- PERCEPTRON

Inflated interest

- ADALINE
- DENDRAL
- ELIZA

First ANN Device - “Stochastic Neural Analog Reinforcement Computer” or SNARC. It was developed in 1951 by Marvin Minsky and Dean Edmonds.

First AI Program - The first AI program is generally recognized as the “Logic Theorist,” developed by Allen Newell and Herbert A. Simon in 1955.

AI Term Was Coined: Dartmouth conference of Artificial Intelligence in 1956.

Perceptron: Is one of the earliest models of a neural network and plays a significant role in the history of artificial intelligence and machine learning. It was developed in 1957 by Frank Rosenblatt.

ADALINE: Another significant ANN precursor - single-layer artificial neural network that uses a linear activation function.

DENDRAL and ELIZA:programs that contributed to the excitement. DENDRAL was an expert system for chemical analysis, and ELIZA was an early natural language processing program.

1st AI Winter. After the Lighthill, J. Artificial Intelligence: A General Survey. Plus, lack of computational power to do anything substantial. Disappointment with researchers led to closing of funding for AI projects.

Expert Systems. New idea opened a new era of AI solutions. ES-based solutions had its own market, and had many applications for real-world problems.

1st AI Winter

- Artificial Intelligence: A General Survey
- Lack of computational power

Expert systems

- XCON Expert System
- Fifth Generation Computer Project
- LISP machines market collapse

2nd AI Winter

- FAILED

The 2nd AI Winter was characterized by a decline in funding, interest, and overall progress in artificial intelligence research, similar to the 1st one.

Desktop computers become faster and more powerful than LISP machines. ⇒ An entire half a billion dollars industry was demolished overnight.

Japan's Fifth Generation Computer Project (create advanced AI-driven computers that could perform tasks like natural language understanding and logical reasoning) outcomes. ⇒ The impressive list of goals had not been met

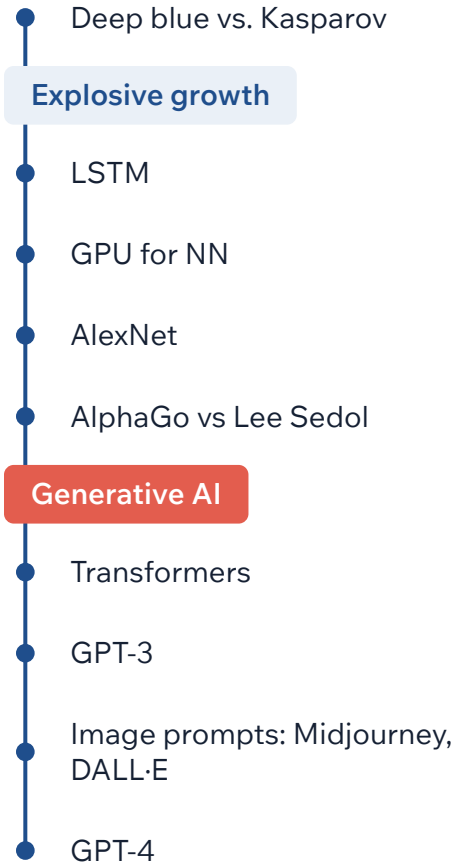
Deep Blue vs. Kasparov (1997): IBM's Deep Blue defeats world chess champion Garry Kasparov, marking a significant AI milestone.

Explosive Growth (2000s - 2010s):

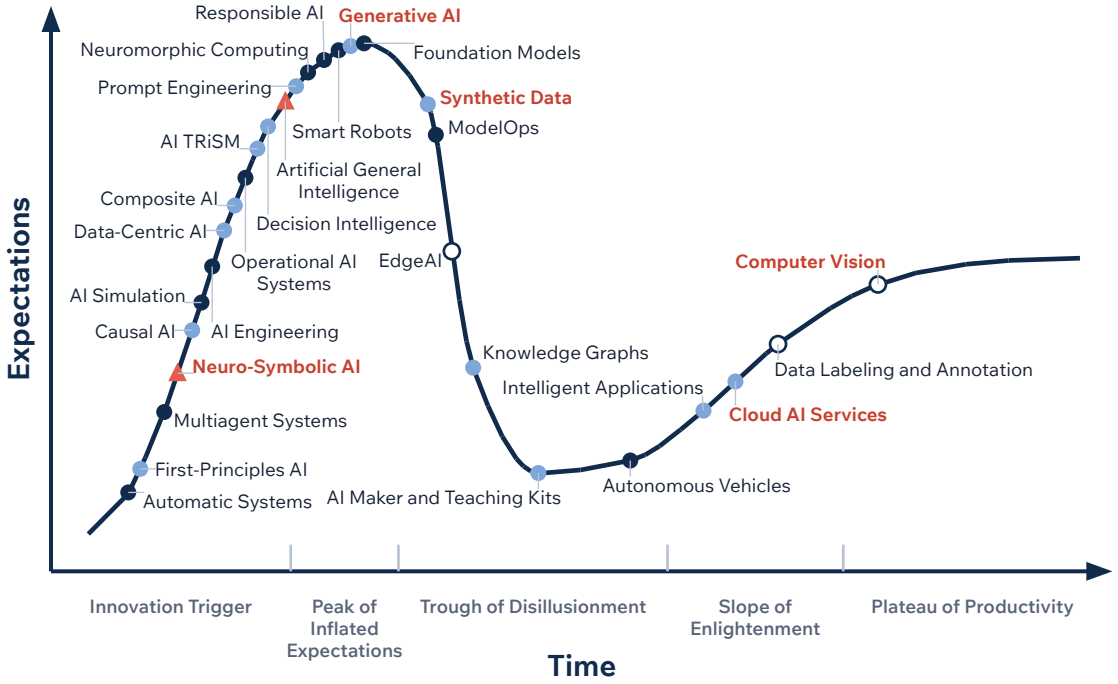
- GPU for Neural Networks: GPUs revolutionize deep learning by providing the computational power needed for training complex models.
- AlexNet (2012): AlexNet's victory in the ImageNet challenge brings deep learning to the forefront of computer vision.
- AlphaGo vs. Lee Sedol (2016): AlphaGo's win over Lee Sedol demonstrates the power of reinforcement learning and deep neural networks.

Generative AI Era (2010s - 2020s):

- Transformers (2017): The Transformer architecture revolutionizes natural language processing.
- GPT-3 (2020): OpenAI's GPT-3 sets new standards in text generation with 175 billion parameters.
- Image Prompts: Models like DALL·E and Midjourney enable the creation of images from text prompts, opening new creative possibilities.



Hype Cycle for AI in 2023



Plateau will be reached:

○ less than 2 years ● 2 to 5 years ● 5 to 10 years ▲ more than 10 years

Source: Gartner, Hype Cycle for Artificial Intelligence, July 2023

Innovation Trigger
▲ **Neuro-Symbolic AI**

Peak of Inflated Expectations
● **Generative AI**

Trough of Disillusionment
● **Synthetic Data**

Slope of Enlightenment
● **Cloud AI Services**

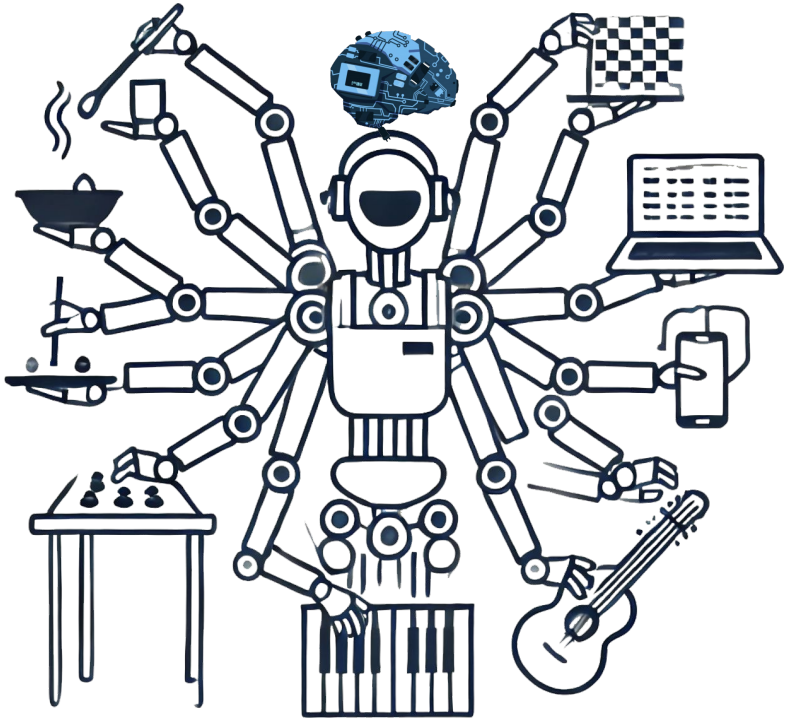
Plateau of Productivity
○ **Computer Vision**

03

Strong vs Weak AI

In fact, the entire history of AI development can be viewed through the lens of trying to understand what kind of AI we want to create. Over time, two main concepts have emerged: strong AI and weak AI.

Strong (General) AI



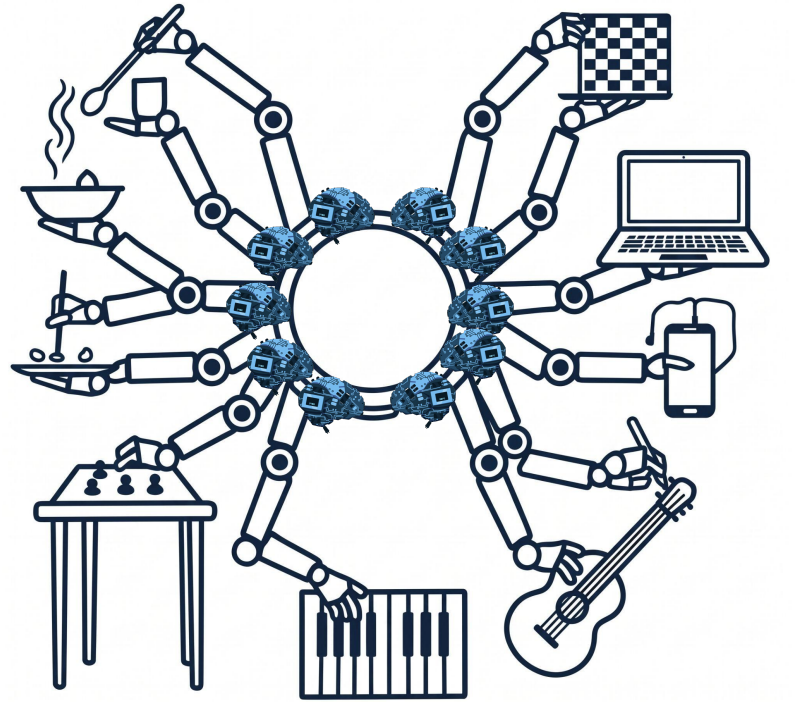
Strong AI, also known as Artificial General Intelligence (AGI), refers to the concept of a machine that possesses the ability to understand, learn, and apply intelligence across a broad range of tasks at a level comparable to or exceeding human capabilities.

The ideas of creating strong AI were most developed in the 1950s and 1960s.

However, by the 1970s and 1980s focus shifted to the development of narrow AI.

Nowadays, discussions about strong artificial intelligence are gaining new momentum due to the success of large language models.

Weak (Nerrow) AI



Narrow AI, also known as Weak AI, refers to the concept of a machine designed to perform a specific task or a narrow range of tasks with high efficiency, without possessing general intelligence or the ability to perform tasks outside its programmed domain.

Since 80s, narrow AI has seen significant advancements, becoming integral to various industries through applications like speech recognition, image processing, and recommendation systems.

Now we live in the world which use narrow AI much, I this is exactly the one AI we will learn in our next lectures.

Different schools of thoughts

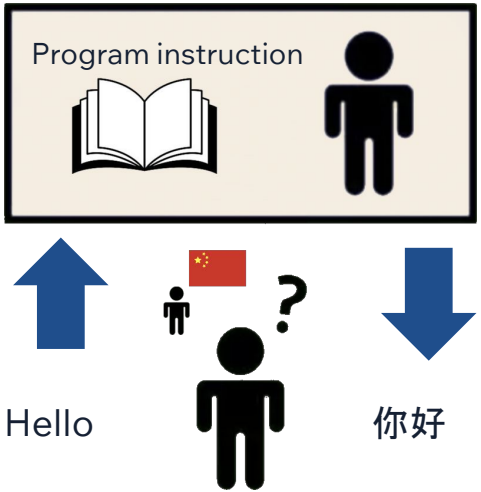
	Dan Dennett & David Chalmers	John Searle	Roger Penrose
			
Strong AI Can a computer fully simulate consciousness?	✓	✗	✗
Weak AI Can a computer fully emulate consciousness?	✓	✓	✗

Chinese room argument



John Searle (philosopher)
1980: «Minds, Brains, and Programs»

Thought experiment proposed by philosopher to challenge the notion of “strong AI” which claims that a computer running a program can have a “mind” and “understand” in the same way humans do.



In the experiment, a person who doesn't know Chinese uses a rule book to match Chinese symbols to produce responses, without understanding the language. Searle argues that, similarly, even if a computer can process language and provide appropriate outputs, it doesn't mean it actually understands the content.

“AI Effect”

Deep Blue beats Garry Kasparov



Only humans can play chess



Major milestone in AI development

Was it even an intelligent task?



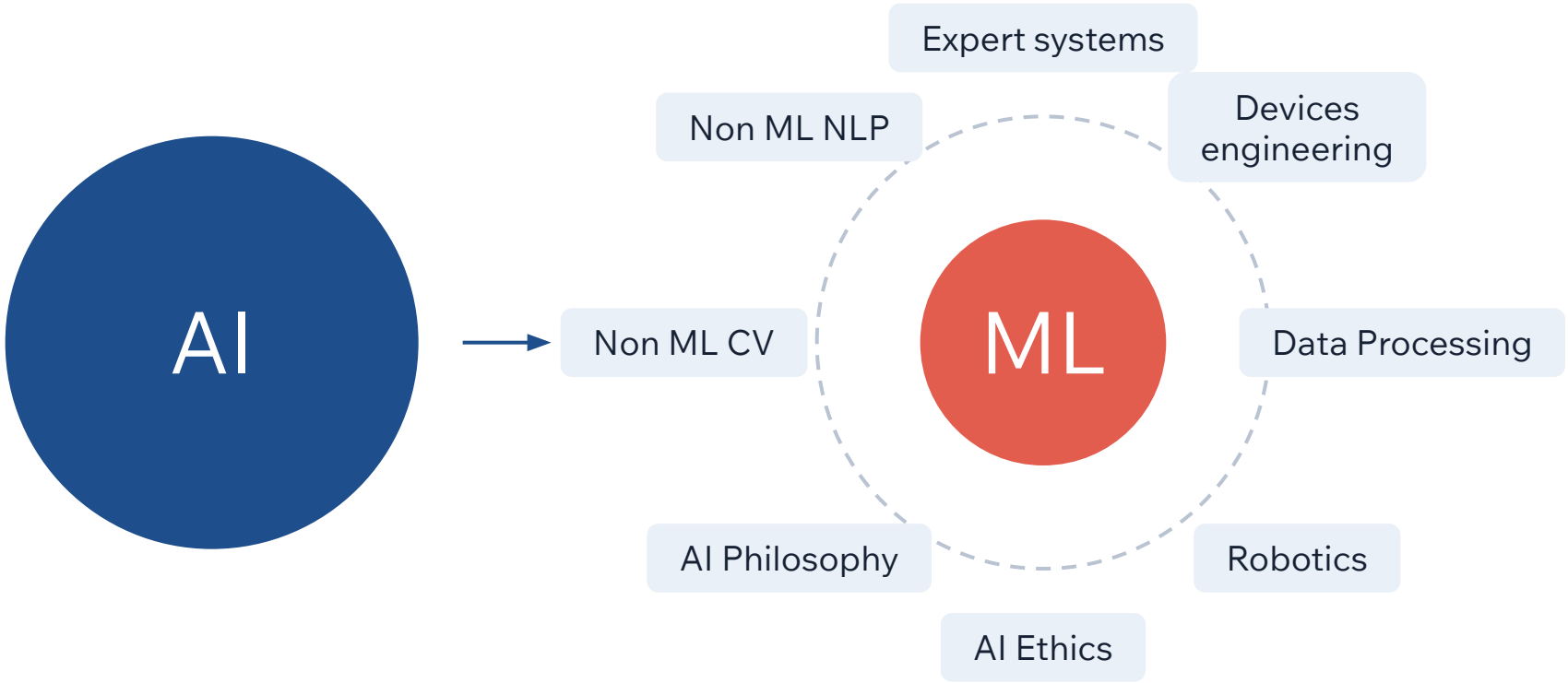
The “AI Effect” refers to the phenomenon where, as AI technology becomes more commonplace and its capabilities are better understood, people tend to dismiss the accomplishments of AI as simple or “not real AI.” Essentially, once an AI system can do something that was previously thought to require human intelligence, that task is no longer considered a benchmark of AI.

04

Machine learning in AI

Machine learning has become a cornerstone in the realm of artificial intelligence today. Let’s dive deeper into what it entails. But first, let’s strip away all the extraneous elements.

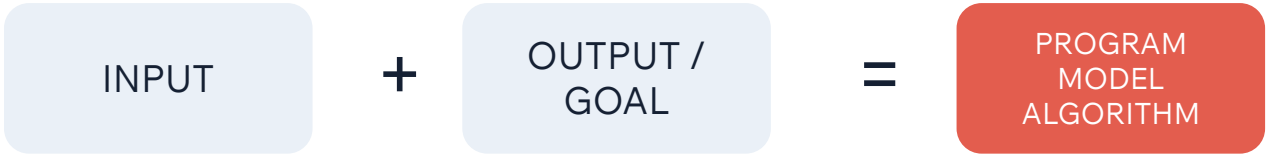
AI structure



ML vs Programming



Programming - Program some predefined dependency between input and output. Understanding of the process comes before the realization and coding.



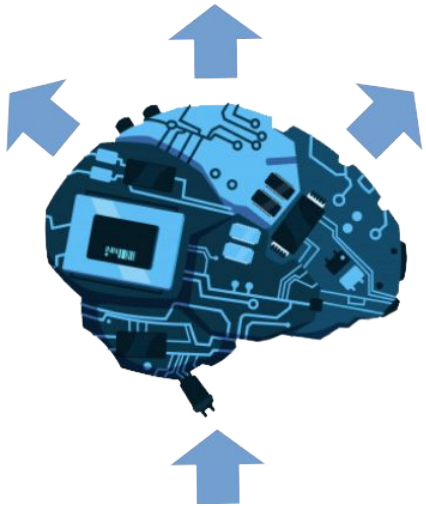
Machine Learning - Develop the algorithm which will define dependency between input and output. In such approach we usually do not understand process before modeling.

Main concept



FIT

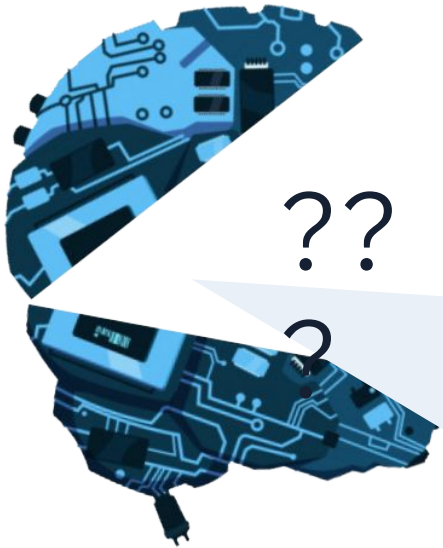
Model learns from data by identifying patterns during training. This phase builds the model's ability to understand and represent the data.



PREDICT

Trained model applies its learned patterns to new data, generating insights or making decisions based on its training.

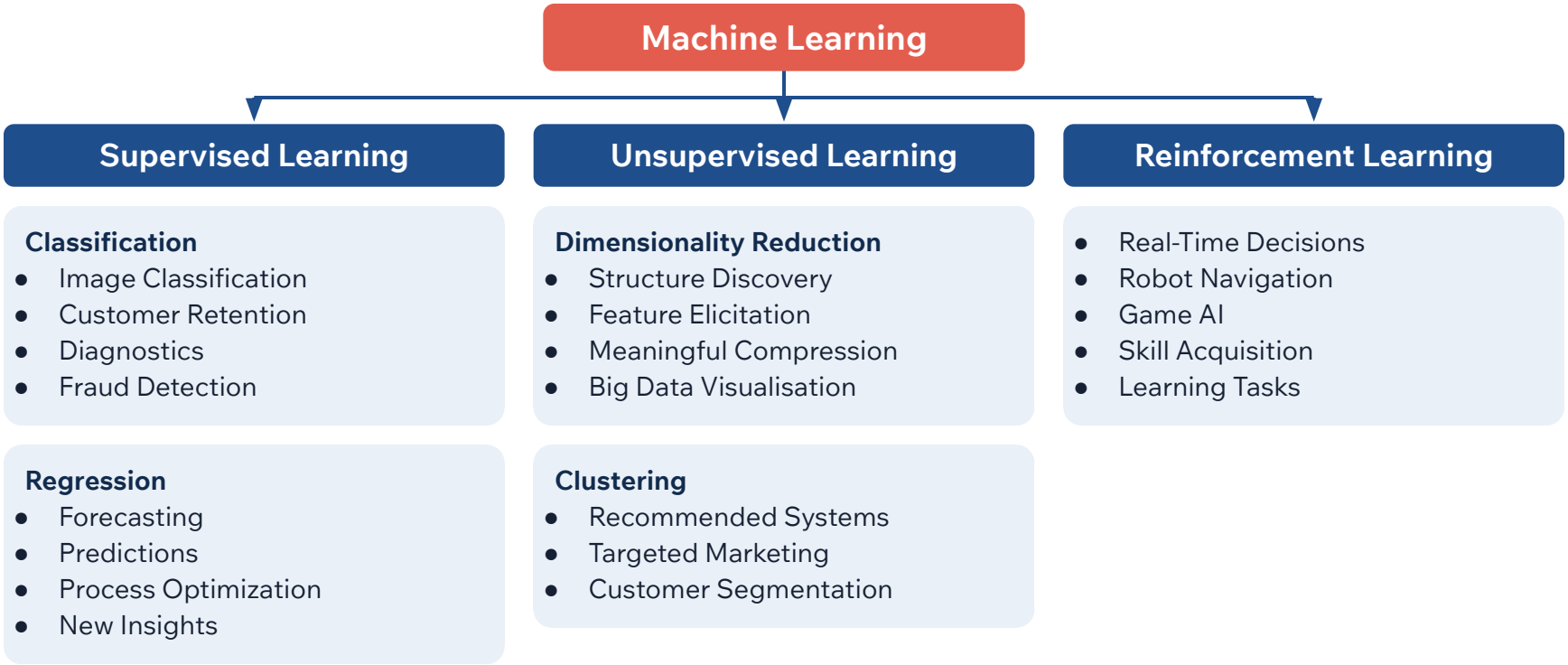
Black box problem



Refers to the lack of transparency in how complex models, especially deep learning algorithms, make decisions.

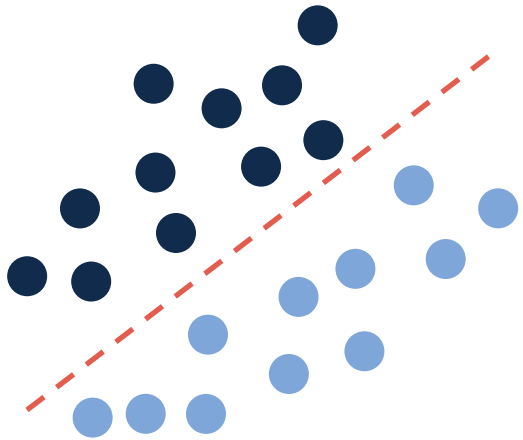
While these models can achieve high accuracy, they often do so without providing clear explanations for their outputs, making it difficult for users to understand or trust the results.

ML Structure

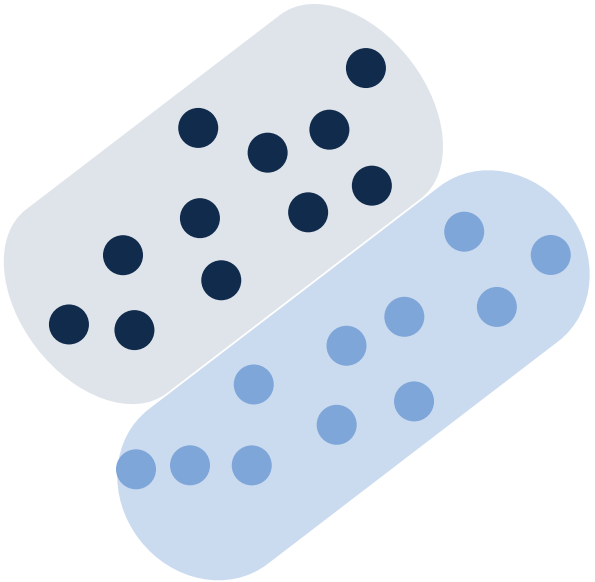


Place of Generative AI

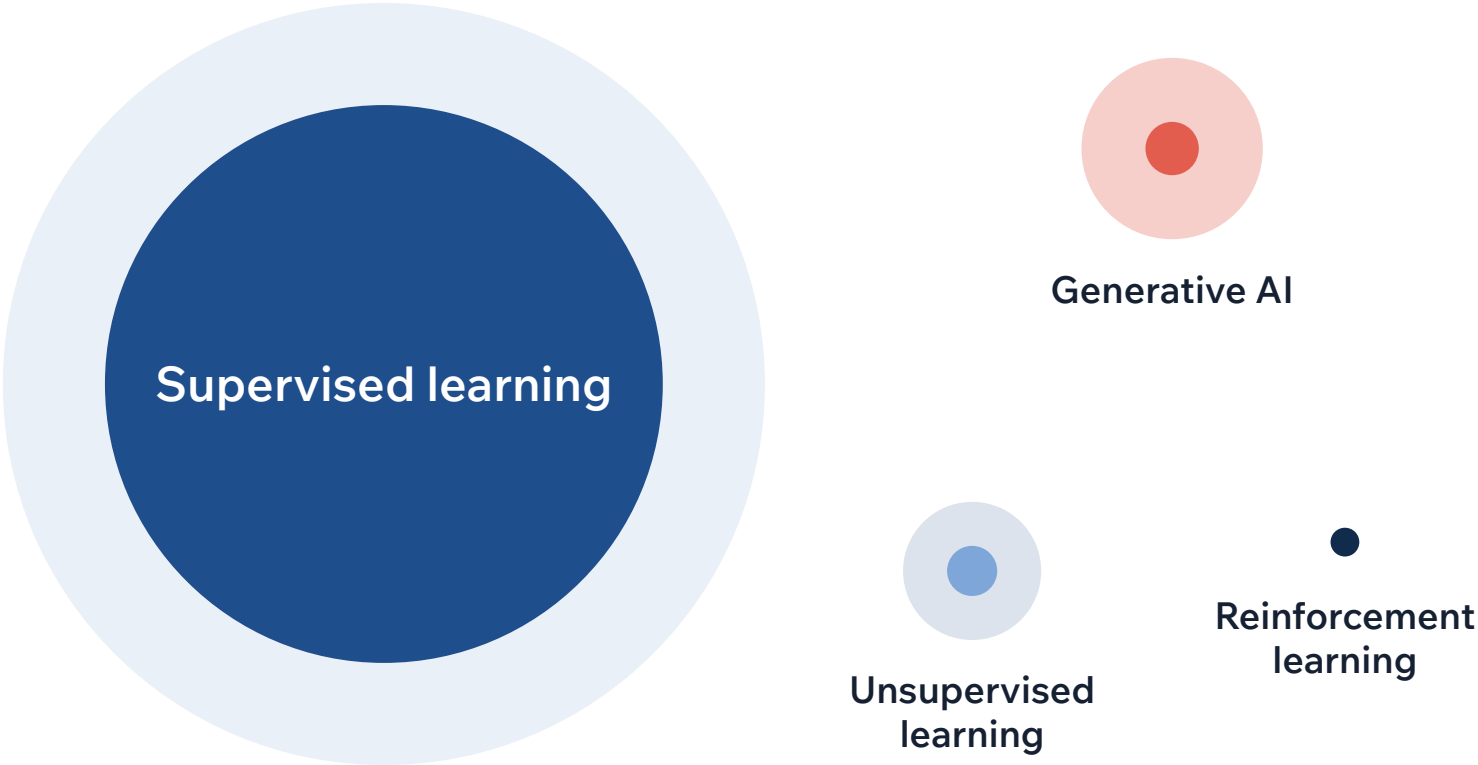
Discriminative AI



Generative AI



Place of Generative AI

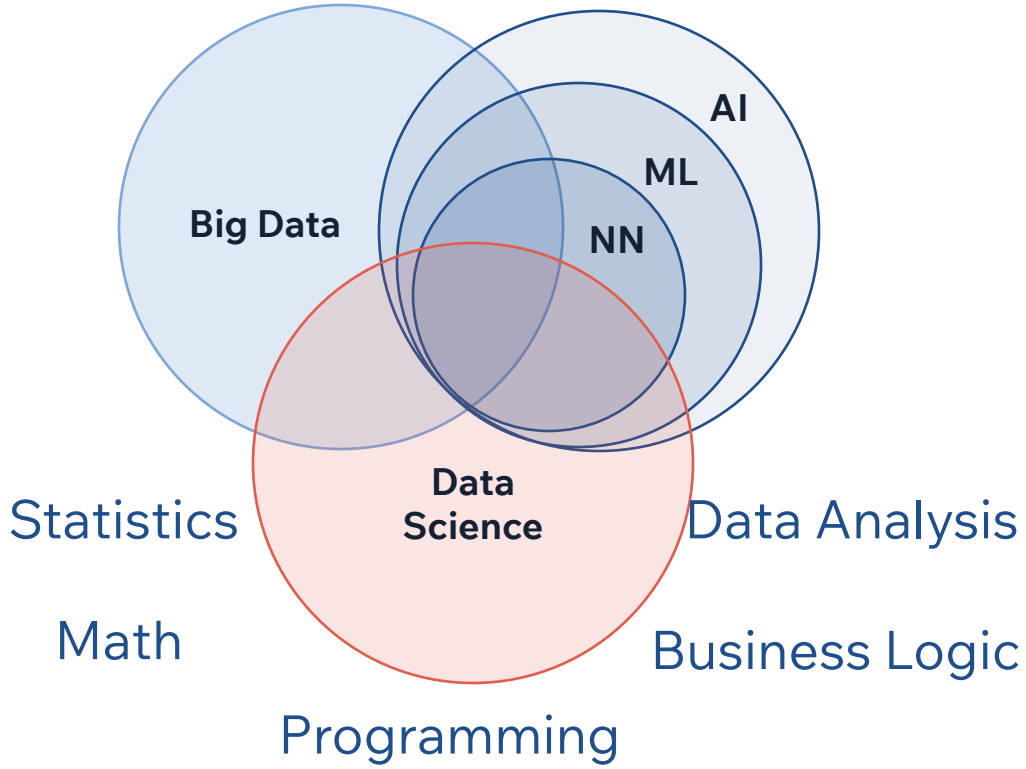


05

Data Science

Ok, but you can ask, why do we speak about AI all this time, but not about Data Science? To clarify this, let’s generalise all the information we have gained and point out the similarity and difference between AI and DS

AI-adjacent fields



DS vs ML

Data Science

Field that determines the processes, systems, and tools needed to transform data into insights to be applied to various industries.

Skills needed:

- Statistics
- Data visualization
- Coding skills (Python/R)
- Machine learning
- SQL/NoSQL
- Data wrangling



Machine learning is part of data science. Its algorithms train on data delivered by data science to “learn.”

Skills needed:

- Math, statistics, and probability
- Comfortable working with data
- Programming skills

Machine Learning

Field of artificial intelligence (AI) that gives machines the human-like capability to learn and adapt through statistical models and algorithms.

Skills needed:

- Programming skills (Python, SQL, Java)
- Statistics and probability
- Prototyping
- Data modeling

DS Stack

Big data stack





Neural networks





Data Operations





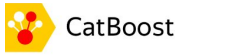


Visualisation



ML Algorithms









Languages





ML general



That's all for today, thank you for your attention.

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